USB POWER 45W S8.4 / SINGLE BOARD, +5V SERIES PROTECTION

S8.4: +5V fast series disconnect (FLT off the latch); +/−12V directly from converters; ceramic banks; two parallel outputs; VPD−powered OV guard; arming from MAIN_OK; MAIN LED only.

USB PD, main eFuse and VIN damper

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A common input switch
VIN damper 1R + 2 x 10u

File: 01_usb_input.kicad_sch

+12 V converter

MAIN_OK
VIN
+12V

TPS62933F
+12 V delivered rail

File: 02_positive_12v.kicad_sch

+5 V converter

MAIN_OK
VIN
P5_RAW

TPS62933F
P5_RAW before disconnect

File: 03_positive_5v.kicad_sch

−12 V inverting converter

+12V
VIN
−12V

LMR38020F inverting buck−boost
−12 V delivered rail

File: 04_negative_converter.kicad_sch

Two output headers − monitors, overvoltage guard and MAIN LED

+12V
−12V
MAIN_LED_OK
P5_RAW
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK
OV_N

Two Eurorack outputs J2 + J3
Monitors, OV guard, +5V eFuse
MAIN LED (only status LED)

File: 06_outputs_status.kicad_sch

FAULT POLICY
Qualification starts at MAIN_OK (converters enabled); persistent bad status after
~0.57 s −> all OFF until RESET. Severe OV via U20 acts even before ARMED.
U23 independently cuts +5V; OVLO alone is auto−recovering. U23 FLT is not latched.
MAIN indicates enabled/status, not precision regulation.

OUTPUTS: +12 / +5 / −12 on J2 and J3 in parallel
Published conservative rating (S8.4, prototype): 2.5 / 2.0 / 0.9 A.
<=45 W total; above 2 A per rail use both headers. U23 ILIM 3.03 A nominal is a threshold, not a rating.

Always−on control − fault latch, overvoltage preset and RESET

MAIN_OK
N12_OK
OV_N
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always−on LDO / retained ARMED + FAULT
Manual RESET; OV_N presets FAULT

File: 05_fault_reset.kicad_sch

Arming from MAIN_OK; U18/U20 on VPD; split +5V banks; see docs/s8−single−board.md
S8.4: single board, +5V series eFuse (FLT off the latch); +/−12V unswitched; MAIN only

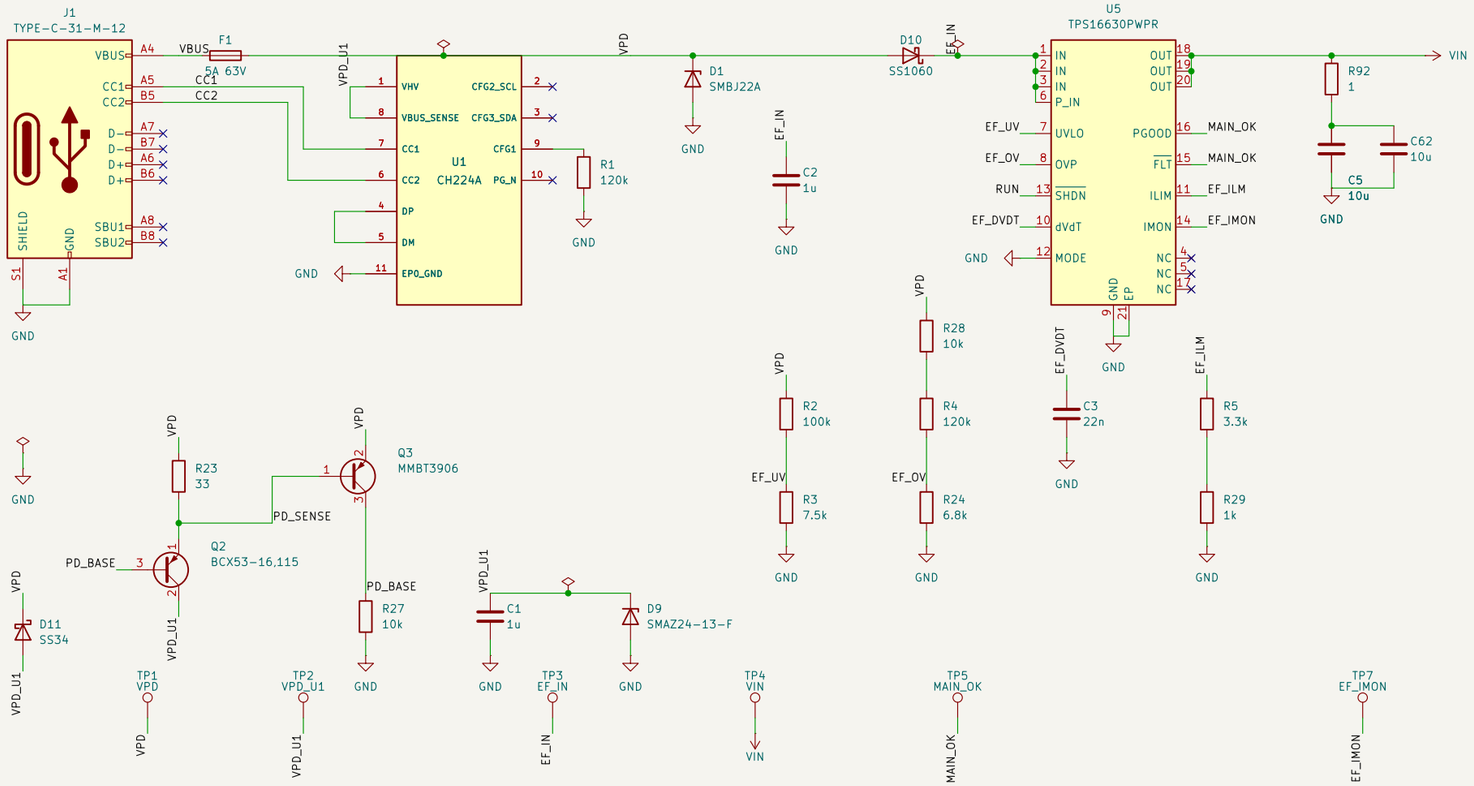
Big Modular

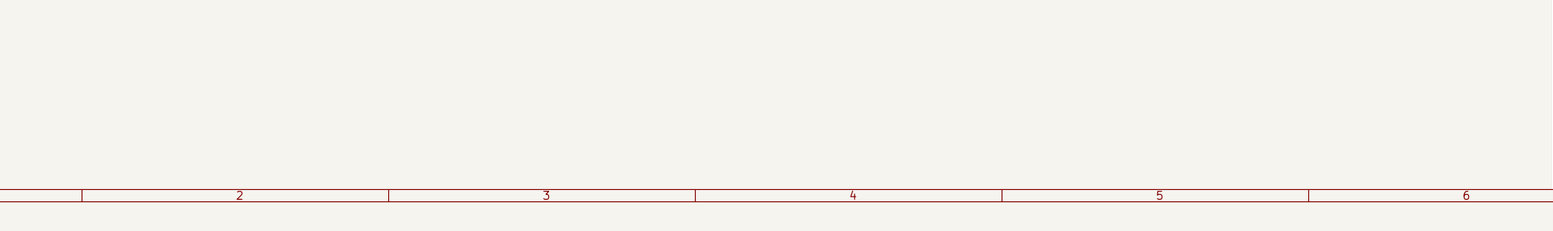
Sheet: /
File: usb−power−45w−single.kicad_sch

Title: USB power S8 single board − wired power flow overview

Size: User	Date:	**Rev: S8.4**
KiCad E.D.A. 10.0.6		Id: 1/7

USB PD, MAIN EFUSE AND VIN DAMPER





20 V PD request: U5 TPS16630 (60 V, 4.186 A nominal) is the common input switch: RUN low turns every rail off.
MAIN_OK = FLT + PGOOD at 3.3 V. R92 1R + C5/C62 2 x 10u damp the all-ceramic VIN bus (replaces the 47u can).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK
RUN
VIN
VPD

S8.4 single board: +5V series eFuse, FLT off the latch, MAIN_OK arming — docs/s8-single-board.md

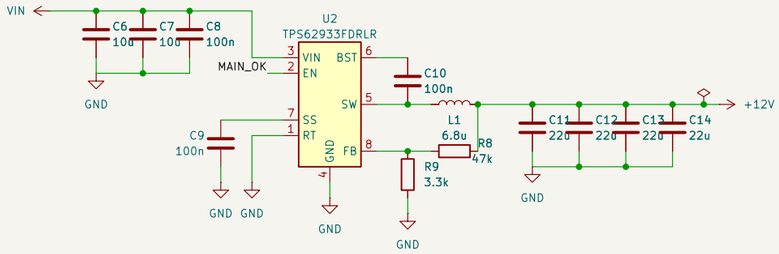
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Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S8.4**
KiCad E.D.A. 10.0.6		Id: 2/7

+12 V CONVERTER





TPS62933F +12 V, unchanged power stage; its output IS the delivered +12V (no rail eFuse).
Soft start C9 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C11–C14 here + C34 (J2) and C59 (J3) on page 06; the 100u can is gone.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇ +12V

MAIN_OK ◇ MAIN_OK

VIN ◇ VIN

S8.4 single board: +5V series eFuse, FLT off the latch, MAIN_OK arming — docs/s8-single-board.md

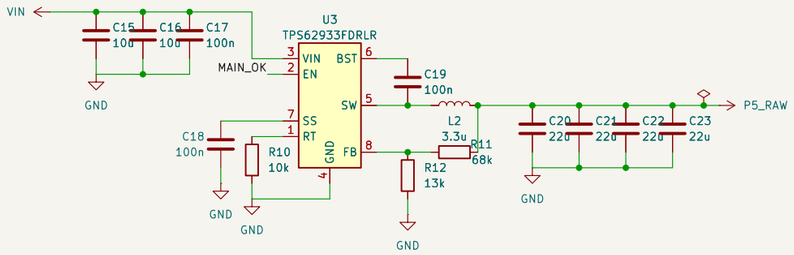
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Sheet: /+12 V converter/
File: 02_positive_12v.kicad_sch

Title: +12 V converter

Size: User	Date:	**Rev: S8.4**
KiCad E.D.A. 10.0.6		Id: 3/7

+5 V CONVERTER





TPS62933F +5 V: P5_RAW before U23. Feedback R11/R12 (0.1 %, S8.4) and C20–C23 remain upstream; never sense after U23.
C69–C72/C75/C76 extend the upstream bank to 10 x 22u. Delivered +5V has C35/C60/C73/C74 after U23.
C18 100n converter soft start; U23 has its own dVdt ramp. Converter/load stability is not bench qualified.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇—— MAIN_OK

P5_RAW ◇—— P5_RAW

VIN ◇—— VIN

S8.4 single board: +5V series eFuse, FLT off the latch, MAIN_OK arming — docs/s8-single-board.md

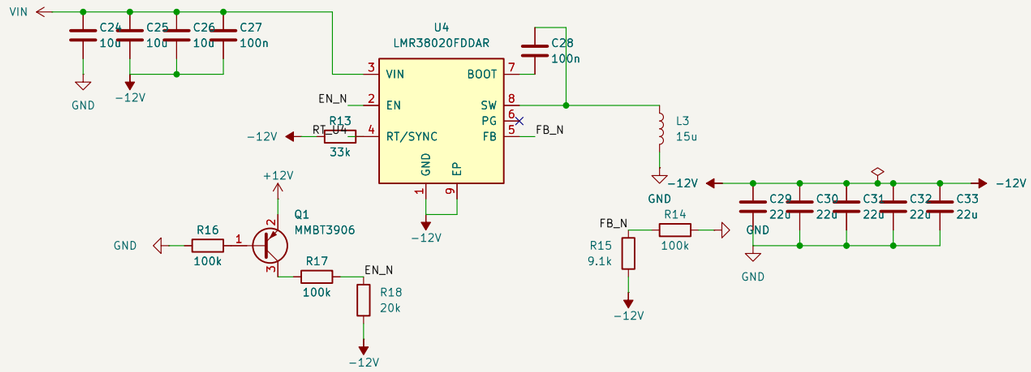
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Sheet: /+5 V converter/
File: 03_positive_5v.kicad_sch

Title: +5 V converter

Size: User	Date:	**Rev: S8.4**
KiCad E.D.A. 10.0.6		Id: 4/7

−12 V INVERTING CONVERTER





LMR38020F inverting buck−boost; IC ground / exposed pad = the delivered −12V (no series switch).
4 ms soft start, 18 ms OCP blanking at start−up. Q1 follows +12V. C29−C33 here + C36 (J2) and C61 (J3).
S8.4: hiccup needs |Vout| < 40 % after the blanking. Rack allowance 0.85 mF (TI 1000 uF COUT guidance; hiccup model
1.08 mF at 15.2 V converter input, 1 A) covers a typical 0.44 mF rack. Bench start−up test still required.

INTER−SHEET INTERFACE (see wired power−flow overview)

+12V ◇ +12V

−12V ◇ −12V

VIN ◇ VIN

S8.4 single board: +5V series eFuse, FLT off the latch, MAIN_OK arming − docs/s8−single−board.md

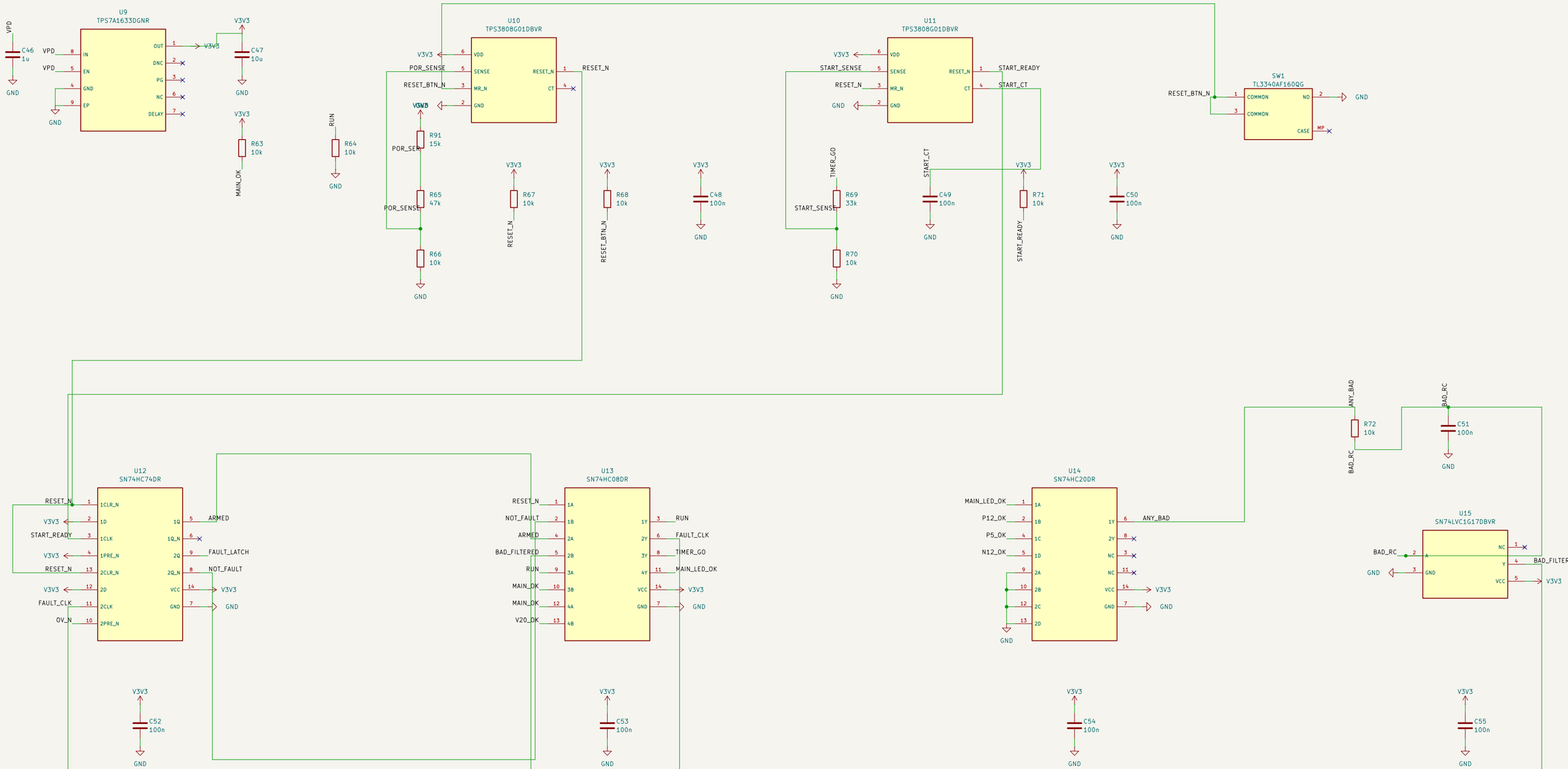
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Sheet: /−12 V inverting converter/
File: 04_negative_converter.kicad_sch

Title: −12 V inverting converter

Size: User	Date:	**Rev: S8.4**
KiCad E.D.A. 10.0.6		Id: 5/7

ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET



TP9 RUN · TP10 ARMED · TP11 FAULT_LATCH · TP12 P12_OK · TP13 P5_OK · TP14 N12_OK · TP15 V20_OK · TP16 V3V3 · TP6 OV_N

RESET clears ARMED and FAULT; RUN is forced low while reset is asserted.
S8.4: TIMER_GO = RUN AND MAIN_OK, the U2/U3 enable; qualification (~0.57 s, 0.31–0.88 s) starts with the converters, so ARMED cannot stay masked while they run. Then any persistent bad status latches every rail OFF until RESET.
OV_N (U20, page 06) presets FAULT at once, also inside the startup window. U14 gate 2 unused (inputs to GND).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_LED_OK · P5_OK · MAIN_OK · RUN · N12_OK · V20_OK · OV_N · V3V3 · P12_OK · VPD

S8.4 single board: +5V series eFuse, FLT off the latch, MAIN_OK arming — docs/s8-single-board.md

Big Modular

Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 05_fault_reset.kicad_sch

Title: Always-on control — fault latch, overvoltage preset and RESET

Size: User	Date:	Rev: S8.4
KiCad E.D.A. 10.0.6		Id: 6/7

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

J2 + J3: two parallel 16-pin outputs, pin 1 = −12V. C34–C36 at J2, C59–C61 at J3 (22u each, replace the 100u cans).
D8 MAIN is the only status LED (rail LEDs removed): input-valid AND main-eFuse status AND RUN. Any bad rail after the
startup window or a U20 guard trip latches every rail off. Short OVLO—only pulses may recover. P12/P5/N12_OK: logic + test pads.
U20 overvoltage guard (S8.1): +12V > 14.88 V, +5V > 5.83 V or −12V < −13.64 V nominal (corners 14.48–15.27 / 5.68–5.98 /
13.27–14.02 V) pulls OV_N -> FAULT -> U5 off. Own reference REF_OV = U21 TLV431B (R100, C67, TP17); U18/U20 LM2901B:
R93–R96, R98, R99 are 0.1 % / 25 ppm. The −12V channel compares MON_OVN with GND (no V3V3 term): D22 holds MON_OVN near
−0.24 V (25 C only, <=0.15 mA) if V3V3 is lost while the rack holds −12V. D21 SMAJ5.0A is a transient clamp, not an OV limit.
S8.3: U18/U20 supplies are always-on VPD: outputs remain 3.3 V logic, U20 +5V guard monitors P5_RAW.
U23 TPS259470A independently disconnects +5V at -5.70 V (1.2 us typical, no maximum specified). D21 moved to P5_RAW.
S8.4: P5_OK = AUXOFF + U18 only: FLT (reverse current/overload/OT) goes to R108/TP21, never to the latch.
Severe raw OV is unmasked through U20. No guaranteed 6 V header peak or prototype qualification is claimed.

U18 LM2901BIDR
REF_GOOD
MON_P5
REF_GOOD
MON_P12
MON_N12
REF_GOOD
REF_GOOD
MON_20
1OUT P5_OK
2OUT P12_OK
3OUT N12_OK
4OUT V20_OK
VCC VPD
GND GND

U20 LM2901BIDR
MON_OV12
REF_OV
MON_OV5
REF_OV
GND
MON_OVN
REF_OV
GND
1OUT OV_N
2OUT
3OUT
4OUT VPD
VCC
GND GND

V3V3 R97 10k
V3V3 R35 10k P12_OK
R44 10k P5_OK
R62 10k N12_OK
+12V R93 110k MON_OV12
P5_RAW R95 37k MON_OV5
R94 10k C63 1n GND
R96 10k C64 1n GND
VPD C65 100n GND
V3V3 R100 2.2k REF_OV
TP17 REF_OV
U21 TLV431BIDBZR GND
C67 10u GND
REF_OV R98 10k
MON_OVN
BAT54SLT1G GND D22
R99 110k C66 1n −12V GND
V3V3 R22 1k
D8 MAIN_LED
MAIN_LED_OK Q6 2N7002
LED_RETURN
+5V R77 33k MON_P5_SER
V3V3 R80 100k MON_N_SER
+12V R75 120k MON_P12
R78 12k MON_P5
R81 10k MON_N12
VPD R83 200k MON_20
V3V3 R85 10k V20_OK
V3V3 R73 3.3k REF_GOOD
R74 1k GND
C56 100n GND
C57 100n GND
R76 10k GND
R79 10k GND
R82 470k −12V
BAT54SLT1G GND D15 V3V3
MON_N12
R84 10k GND
LED_RETURN
RUN Q10 2N7002 GND

P5_RAW
C69 22u C70 22u C71 22u C72 22u C75 22u C76 22u GND
TP18 P5_RAW
P5_RAW R107 510 P5_OV_TOP
R101 37k P5_OVLO
R102 10k GND
U23 TPS259470ARPWR
P5_RAW IN
RUN EN/UVLO
P5_OVLO OVLO
ITIMER
GND
DVDT
OUT +5V
AUXOFF P5_OK
FLT P5_FLT_N
ILM P5_ILM
P5_DVDT
C68 10n GND
P5_ILM R103 1k P5_ILM_SER
R104 100 GND
P5_RAW D21 SMAJ5.0A
V3V3 R108 10k P5_FLT_N
TP21 P5_FLT_N
+5V C73 22u C74 22u D23 SS34 GND

J2 IDC 2x8 box header
J3 IDC 2x8 box header
−12V GND +12V +5V
C34 22u C35 22u C36 22u
C59 22u C60 22u C61 22u

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V +12V
−12V −12V
MAIN_LED_OK MAIN_LED_OK
N12_OK N12_OK
OV_N OV_N
P12_OK P12_OK
P5_OK P5_OK
P5_RAW P5_RAW
RUN RUN
V20_OK V20_OK
V3V3 V3V3
VPD VPD

S8.4 single board: +5V series eFuse, FLT off the latch, MAIN_OK arming — docs/s8-single-board.md
Big Modular
Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 06_outputs_status.kicad_sch
Title: Two output headers — monitors, overvoltage guard and MAIN LED

Size: User	Date:	**Rev: S8.4**
KiCad E.D.A. 10.0.6		Id: 7/7